Motor Imagery Recognition Using Filter Bank Common Spatial Pattern and Recurrent Neural Networks

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***Abstract* — A Brain-Computer Interface (BCI) technology allows a person to communicate with hardware without involving physical movement. The instrument that connects brain commands to device movements is called an intermediate, one of which is an Electroencephalogram (EEG). EEG captures brain signals by placing electrodes on the scalp and translating the recorded signals to external commands. In BCI, one of the variables that can be used is motor imagery (MI), where a person can control an external device by imagining its movements. EEG signal processing is important in BCI performance. There are feature extraction and classification. This paper proposed the Common Spatial Pattern (FBCSP) technique for feature extraction and Recurrent Neural Network (RNN) to recognize BCI competition data sets. FBCSP is effective in MI recognition because it extracts frequency and spatial features. RNN is used for the classification by considering the data sequence of each spatial feature of FBCSP extraction. The experiment result showed that of the 9 dataset subjects, the accuracy of each subject is 94.25%, 87.36%, 94.25%, 90.80%, 85.06%, 89.66%, 95.40%, 95.40%, and 93.10, with an average accuracy of 91.69%. If all subject data is combined, the accuracy is 89.97%.**

Keywords— Brain-Computer Interface, Motor Imagery, Electroencephalogram, Common Spatial Pattern, Recurrent Neural Network

I. INTRODUCTION

Brain-computer interface (BCI) is a technology that allows a person to communicate with external devices through brain signals without having to involve physical motion activity [1]. One of the important variables in BCI is motor imagery (MI), which allows one to control an external device by imagining the body's motion[2]. BCI uses various intermediary methods, one of the most commonly used being the electroencephalogram (EEG).

The electroencephalogram (EEG) plays an important role in the BCI system and becomes one of the main methods for obtaining brain signals. EEG uses electrodes attached to the scalp to measure the electrical activity of the brain, which is then translated into signals that the machine can receive [3]. EEG signals are grouped into several frequency waves, such as Delta (1-4 Hz), Theta (4-8 Hz), Alpha (8-13 Hz), Beta (13-30 Hz), and Gamma. (30-50 Hz) [4], each showing different brain activity. In the context of motor imagery, Mu waves (8-12 Hz) and Beta waves (13-30 Hz) are represented in EEG signals [5.] These are known as event-related desynchronization and event-released synchronization (ERD/ERS), and they have played important roles in the BCI research.

Some researchers utilized EEG for various purposes, including robot control, medical diagnosis, epilepsy detection, automated car navigation, and drone operation. Interpreting EEG signals related to motor imagery allows for the integration of human intelligence into machines, thereby enhancing the strength and effectiveness of the BCI [6].

Some previous studies have applied several computing models, such as Recurrent Spatio-Temporal Neural Network (RSTNN), to identify subjects in BCI Competition IV-IIb data using MI variables [7], in other studies, there has been an increase in an average accuracy of 11.32%, 3.87%, and 1.37% when using the Common Spatial Pattern (CSP)-Support Vector Machine (SVM) method compared to R-CSP, FBCSP, and SBRCSP[8] and Deep learning, Convolutional Neural Networks (CNN) have been successfully applied to 2D data such as images. However, EEG data, recorded from the scalp using a set of electrodes, involves complex spatial and temporal characteristics. Therefore, applying deep learning to EEG data is somewhat more challenging due to the low signal-to-noise ratio [4], [6].

Common Spatial Pattern (CSP) [9] is one of the effective and popular signal processing methods in the Brain-Computer Interface (BCI) that uses EEG [10]. This algorithm creates a spatial filter to maximize the variance difference between the two classes of EEG signals to extract features associated with Event-related desynchronization (ERD) and Event-related synchronization (ERS) [11]. CSP utilizes the entire EEG channel available in the feature extraction process but has a weakness in assuming that all EEG signals in the channel have correlations and relationships with each other, making it difficult to distinguish between EEG signs related to noise signals [10].Therefore, this research development of CSP, namely the Common Bank Spatial Pattern Filter (FBCSP) [12], which considers the differences between the signal's spatial and EEG frequency features. FBCSP has additional bank filters to solve the correlation between spatial and frequency problems encountered by the CSP algorithm. FBC SP helps distinguish between EEG signals associated with brain activity and non-related noise signals. Therefore, it is regarded as an effective development of the Common Spatial Pattern (CSP) in EEG-based BCI signal processing, and it applies the bank filter and the common spatial pattern to convert the EEG signal into a more useful feature to move on to the modeling and classification.

A recurrent neural network (RNN) is a neural network that can process signals dynamically and consider information from previous times. RNN is very useful in processing EEG signals that have complex and variable time characteristics.

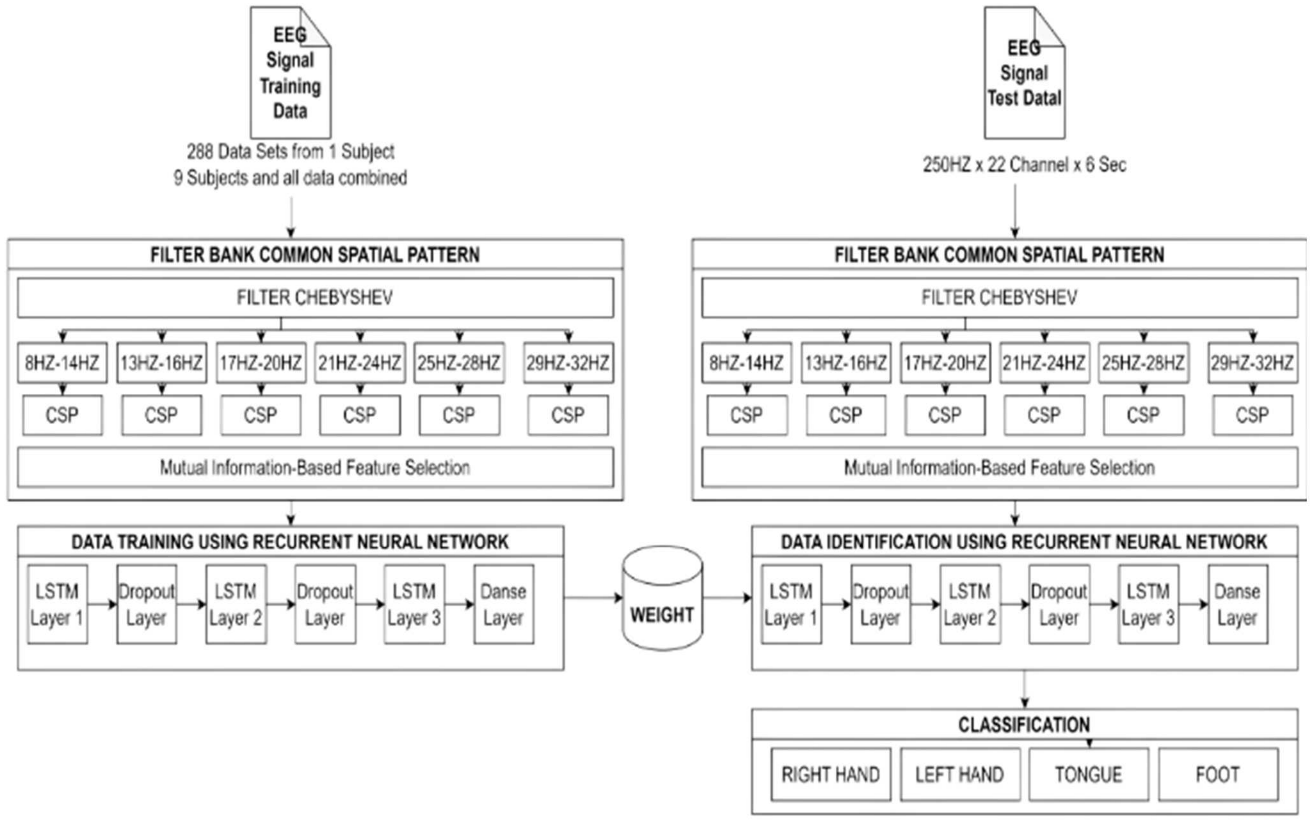


Fig. 1. Filter Bank Common Spatial Pattern and Recurrent Neural Network Model

Using the RNN architecture, EEG-based BCI systems can obtain information about brain activity patterns related to EEG signs and extract features related to Event-related desynchronization (ERD) and Event-Related Synchronisation (ERS). RNN can study complex brain push active patterns and account for historical information, thus facilitating the EEG signal processing process and strengthening the performance of EEG-based BCI [13]. It enables it to improve performance in BCI-MI classification.

This research used data from BCI Competition IV-IIa [14] is one of the sources of EEG data that can be used in BCI and has been extensively studied in this field, consisting of 22 channels. The common spatial pattern bank filter (FBCSP) and recurrent neural network (RNN) computing models are also used to process EEG signals. FBCSP is a signal processing method developed from the Common Space Pattern and has a bank filter that helps consider the correlation between frequency and spatial in EEG feature extraction. RNN, on the other hand, is a deep learning model that is specifically applied to handle sequential data, such as EEG data, and can capture dynamic patterns to classify that consist of four classes: right hand, left hand, tongue, and foot.

II. METHOD

This study has three stages: data acquisition, feature extraction using FBCSP, and RNN for identification systems on BCI-MI. Fig. 1 shows the design diagram of the classification system.

A. Dataset

EEG data obtained using BCI Competition IV-2a [14]. It shows 11 events that capture MI, as shown in Table 1.

TABLE I. EVENTS OF DATASET

Event	Type	Description
276	0x0114	Idling EEG (eyes open)
277	0x0115	Idling EEG (eyes closed)
768	0x0300	Start of a trial
769	0x0301	Cue onset left (class 1)
770	0x0302	Cue onset right (class 2)
771	0x0303	Cue onset foot (class 3)
772	0x0304	Cue onset tongue (class 4)
783	0x030F	Cue unknown
1023	0x03FF	Rejected trial
1072	0x0430	Eye movements
37266	0x7FFE	Start of a new run

There are 25 channels, 22 EEG channels, and 3 EOG channels, nine subjects, and 288 experiments in one subject. However, in this study, only 22 EEG channels were used.

B. Filter Bank Common Spatial Pattern

A Band Pass filter is a signal filter that allows signals in a frequency band. A band Pass filter combines the High Pass and Low Pass filters in one filter. One of them is the Chebyshev, which uses (1)

$$G_{\eta}(\omega) = |H_{\eta}(j\omega)| = \frac{1}{\sqrt{1 + \varepsilon^{2} T_{\eta}^{2}(\omega/\omega_{0})}} \quad (1)$$

ε = ripple factor,

ω0 = cutoff frequency

Tn = Chebyshev polynomial of the n

After that, the function (ω_{pm}) of the Chebyshev filter function is zero for the function denominator. Using the complex frequency s, this happens with (2)

$$1 + \varepsilon^2 T_n^2(-js) = 0 \quad (2)$$

Defining $-js = \cos(0)$ and using Chebyshev's polynomial, trigonometric definition produces like on the (3)

$$1 + \varepsilon^2 T_n^2(\cos(0)) = 1 + \varepsilon^2 \cos^2(n0) = 0 \quad (3)$$

Resolve for 0 using (4)

$$0 = \frac{1}{n} across(\frac{\pm j}{\mathcal{E}}) + \frac{m\pi}{n} \quad (4)$$

where some values of the arc cosinus function are made explicit using the index of the integer m, the Chebyshev amplification function pole is:

$$\mathcal{S}_{pm} = j\cos(0) \quad (5)$$

$$= j\cos\left(\frac{1}{n}arccos\left(\frac{\pm j}{\varepsilon}\right) + \frac{m\pi}{n}\right)$$

Using the properties of trigonometric and hyperbolic functions, this can be written in a complex form explicitly like (6)

$$\mathcal{S}_{pm}^{\pm} = \pm\sinh\left(\frac{1}{n}arsinh\left(\frac{1}{n}\right)\right)\sin(0_m) \quad (6)$$

$$\pm j\cosh\left(\frac{1}{n}arsinh\left(\frac{1}{n}\right)\right)\sin(0_m)$$

Where m = 1,2,…,n and (7)

$$0_m = \frac{\pi}{2}\frac{2m-1}{n} \quad (7)$$

The rest of it has a transfer function using (8)

$$H(s) = \frac{1}{2^{n-1}\varepsilon}\Pi_{m=1}^{n}\frac{1}{8 - 8_{pm}} \quad (8)$$

Common Spatial Pattern (CSP) is a technique used to analyze multi-channel signals such as electroencephalogram signals (EEGs). The most striking signal patterns associated with each condition are found and combined into a spatial pattern that, in this case, is used to extract features in two classes, namely Event-Related Desynchronization (ERD) and Event-related Synchronisation (ERS), on EEG data.

The CSP algorithm involves the following steps:

1. First, count the spatial covariance of each EEG channel during one experiment.
2. Second, create a spatial filter that can separate the features between classes.
3. Finally, the filter is applied to the EEG signal to extract the useful space features for classification.

The E matrix representing EEG data gives the sample per channel of the EEG measurement performed during one experiment. N is the number of channels, and T is the number of sample measurements per channel (9).

$$E = \frac{xx^T}{trace(xx^T)} \quad (9)$$

A spatially filtered EEG signal Z can be given as Z=WE, where W is the CSP projection matrix. The W line contains a stationary spatial filter, and the W-1 column is a common spatial pattern (10).

$$Z_{b,i} = \mathcal{W}_b^T E_{b,i} (10)$$

Here, E represents the tuberculosis matrix corresponding to EEG measurement data from a single experiment. N denotes the number of channels, while T refers to the total number of measurement samples per channel. W stands for the CSP projection matrix, where the rows of W serve as stationary spatial filters, and the columns of W-1 represent the general spatial patterns.

After obtaining the EEG signal filtered with the bank filter, perform the feature calculation using the Common Spatial Pattern (CSP). First, decompose the filtered EEG signals into 2N components, with N being the number of filters in a bank filter.

Next, calculate the CSP transform matrix using (11):

$$W = [B1, B2, \ldots, BN] \quad (11)$$

where B1 to BN is the transformation matrix for each component resulting from the filtered EEG signal decomposition.

With the CSP transformation matrices that have been calculated, the feature can be computed using (12).

$$\log(var(ZW1)), \log(var(ZW2)), . \log(var(ZWN-1)) \quad (12)$$

Z is the EEG signal filtered with a bank filter, and W1 to WN-1 is the CSP transform matrix for each component, except for the last component. The log variance of the filtered and modified signal with this transformation matrix is then used as a feature in MI classification. The number of features produced in FBCSP depends on the number of filters in the bank filter. If there is an N filter in bank filters, then the N-1 feature used for MI classifications will be produced.

The final result of the CSP algorithm is a W projection matrix and a spatially filtered EEG signal Z. The resulting Z signal contains spatial feature information useful for distinguishing between two classes of EEG. This information can be used to classify well using statistical analysis and machine learning methods.

C. Recurrent Neural Network

In Motor Imagery (MI) identification studies, the LSTM architecture is frequently utilized due to its superior performance in managing overfitting and preserving past information. LSTM operates through four key processes within each input neuron or gate unit. The first process, known as the forget gate, determines whether to retain or discard previous data. The second process, the input gate, involves two steps: one similar to the forget gate and another for storing new information. Next, the cell gate combines the outcomes of the forget and input gates to compute the overall value. Finally, the output gate determines which information to output and stores it in the memory cell

III. RESULT AND DISCUSSION

Experiments used FBCSP and RNN from the BCI competition IV dataset recorded from 9 subjects. Each subject's weight correction technique, namely Adamax and SGD, is compared to its average. The study's results are compared with previous studies using the same dataset. The accuracy variable and Loss value show performance.

A. Comparision of the Optimization Model

Several techniques are used to perform weight correction on learning and identification, including AdaMax and SGD, as shown in Table II, with a learning rate 0.0001.

TABLE II. ACCURACY AND LOSS

Subject	Adamax		SGD	
	Accuracy (%)	Loss	Accuracy (%)	Loss
1	94.25	0.0249	89.66	0.0438
2	87.36	0.0530	66.67	0.1020
3	94.25	0.0192	88.51	0.0444
4	90.80	0.0375	81.61	0.0724
5	85.06	0.0609	79.31	0.0744
6	89.66	0.0439	68.97	0.1043
7	95.40	0.0156	90.80	0.0369
8	95.40	0.0208	86.21	0.0493
9	93.10	0.0304	83.91	0.0395
Average	91.69	0.0340	81.73	0.0630
All	89.97	0.0398	68.38	0.1109

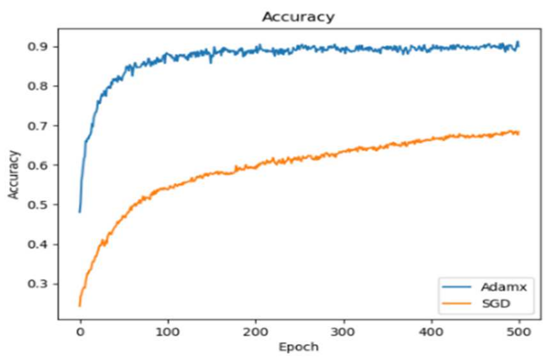


Fig. 2. Comparison Accuracy using Adamax and SGD technique

The Adamax correction technique has an average accuracy of 91.69% per subject compared to the SGD optimization, which only has an average accuracy of 81.73%. The average loss value of Adamax optimization is lower at 0.0340 compared to SGD, which has an average loss value of 0.0630. Adamax performs better than SGD because it is more adaptive, stable, and efficient in handling gradients, especially large noise and high data variation. In addition to using deep learning, it can handle complex parameters. Fig. 2 (Accuracy) and Fig. 3 (Loss) also show how the Adamax technique works.

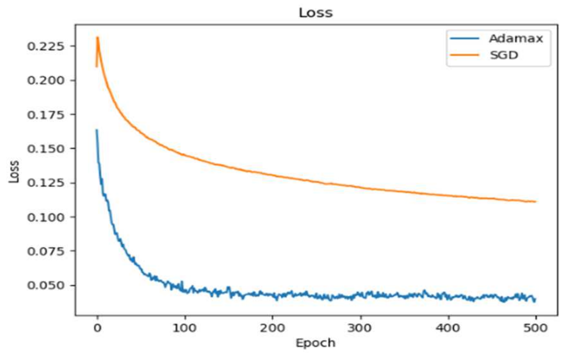


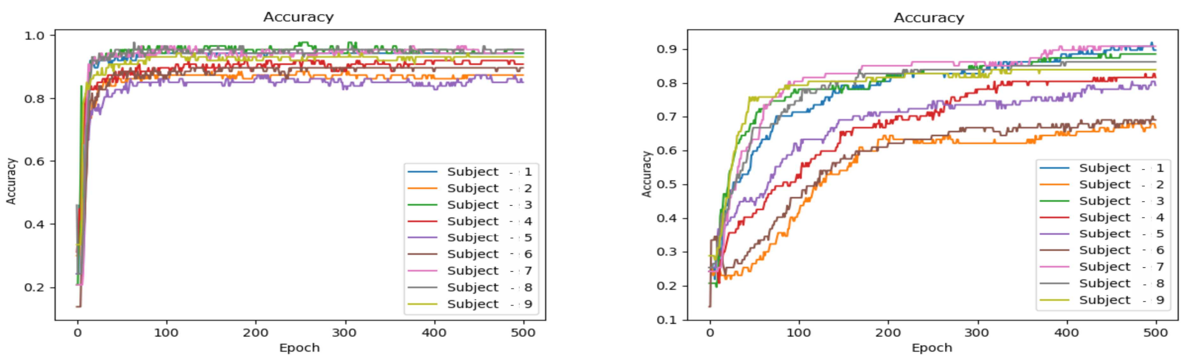
Fig. 3. Comparison Accuracy using Adamax and SGD technique

As shown in Fig. 2 and Fig. 3, the results of the Adamax and SGD optimization models differ significantly. The Adamax optimization model achieves a higher and more stable average accuracy across each data set than the SGD optimization model.

B. Performance of Each Subject

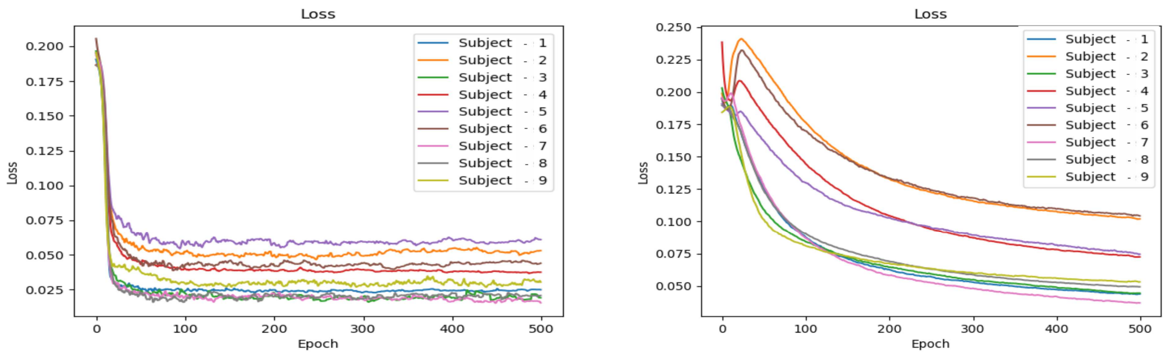
The BCI competition dataset has high subject variability, while the number of subjects is limited. It is also due to the dominance of differences in brain structure and anatomy when brain function is high. Therefore, each subject's performance was tested and compared with previous studies using the same dataset.

Table II describes each subject's performance. The variability of SGD performance is much greater than that of Adamax, which also shows the ability of the Adamax technique to handle it. Likewise, the lower each subject's performance, the greater the difference in the skills of the two optimizers. This phenomenon is shown in Fig. 4 for the accuracy of (a) Adamax and (b) SGD, while Fig. 5 shows the Losses of (a) Adamax and (b) SGD.



(a) (b)

Fig. 4. Accuracy of each subject (a) Adamax (b) SGD



(a) (b)

Fig. 5. Loss of each subject (a) Adamax (b) SGD

TABLE III. ACCURACY USING SEVERAL METHODS

Subject	Propose Methode	Ovo-CSP [10]	CSP-MRCP [15]	FBCSP [16]	FBCSP-SVM [17]	CW-CNN [18]	RBM-tSNE-SVM [17]
1	94.25	90.04	80.0	76.00	82.29	86.11	86.60
2	87.36	59.36	100.0	56.50	60.42	60.76	61.26
3	94.25	86.08	68.3	81.25	82.99	86.81	87.27
4	90.80	71.05	73.3	61.00	72.57	67.36	75.20
5	85.06	58.33	88.3	55.00	60.07	62.50	64.54
6	89.66	54.42	68.3	45.25	44.10	45.14	65.90
7	95.40	91.34	65.0	82.75	86.11	90.63	83.78
8	95.40	85.24	78.3	81.25	77.08	81.25	89.90
9	93.10	79.17	86.7	70.75	75.00	77.08	92.07
Average	91.69	75.00	78.3	67.75	71.18	73.07	78.50

As shown in Fig. 4, the results show a significant difference between the Adamax and SGD optimization models. The Adamax optimization model achieves a higher and more stable average accuracy across each data set than the SGD optimization model. In Fig. 5, the Loss produced by the Adamax optimization model is significantly lower and more stable than that produced by the SGD optimization model across each dataset.

C. Comparison with previous methods

Table III compares previous methods using the same data set (BCI Competition IV). There are Ovo-CSP, CSP-MRCP, FBCSP, FBCSP-SVM, CW-CNN and RBM-tSNE-SVM.

It can be seen that the use of Filter Bank Common Spatial Patterns and Recurrent Neural Networks generally improves the performance of previous studies using the same dataset: One-vs-One Common Spatial Pattern (Ovo-CSP), Common Spatial Pattern- Multi-Resolution Common Spatial Pattern (CSP-MRCSP), Filter Bank Common Spatial Pattern (FBCSP), Filter Bank Common Spatial Pattern- Support Vector Machine (FBCSP-SVM), Convolutional Wavelet Convolutional Neural Network (CW-CNN), and Restricted Boltzmann Machine - t-Distributed Stochastic Neighbor Embedding- Support Vector Machine (RBM-tSNE-SVM). This is indicated by higher accuracy and lower variability. FBCSP's captures frequency information from motor imagery. By using FBCSP, the model can capture more relevant features from various frequencies, thereby improving the model's ability to recognize different patterns in EEG signals. RNN uses memory settings such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), which are very good at capturing temporal or sequential information in data. EEG is a data type with a strong temporal component, where brain activity patterns change over time. RNNs can retain information from previous times to help predict thecurrent state, which is useful for understanding EEG signals. The FBCSP and RNN approaches also help reduce inter-subject and inter-session variability in EEG signals. By capturing more complete features from multiple frequencies and considering temporal information, the models become more robust to individual differences in EEG signals. It leads to more consistent performance across subjects and sessions.

IV. CONCLUSION

The BCI competition dataset has quite high variability, so it is necessary to test each subject's classification performance. The Common Spatial Pattern Bank Filter method, used for feature extraction and classification using Recurrent Neural Networks (RNN), can classify motor imagery with an accuracy of 85.0-96.40% for each subject and an average of 91.69%. The ability to map a fixed frequency band such as FBCSP significantly improves performance.

In addition, the selection of weight correction techniques, often called optimization models, is very important in the performance of accuracy, stability, and generalization, as shown by the decrease in the variability of recognition for each subject. Experiments also show that the lower the classification performance of a particular subject, the higher the variability. EEG signals are inherently noisy and can exhibit significant variability due to individual differences in brain anatomy, mental state, and external factors like muscle artifacts or environmental noise. When a subject's EEG data is highly variable, it becomes more challenging for classification algorithms to identify consistent patterns, leading to lower performance.

Further research is needed for statistical machine learning analysis on highly variable datasets and limited subjects, including cross-validation of the relationship between each subject.

REFERENCE

[1] P. O. de Paula, T. B. da Silva Costa, R. R. de Faissol Attux, and D. G. Fantinato, "Classification of image encoded SSVEP-based EEG signals using Convolutional Neural Networks," *Expert Systems with Applications*, vol. 214, no. July 2021, p. 119096, 2023, doi: 10.1016/j.eswa.2022.119096.

[2] R. Ouyang, Z. Jin, S. Tang, C. Fan, and X. Wu, "Low-quality training data detection method of EEG signals for motor imagery BCI system," *Journal of Neuroscience Methods*, vol. 376, no. March, p. 109607, 2022, doi: 10.1016/j.jneumeth.2022.109607.

[3] D. T. Rajesh Kumar, U. S. B. K. Mahalaxmi, R. MM, and D. D. Bhatt, "Optimization enabled deep residual neural network for motor imagery EEG signal classification," *Biomedical Signal Processing and Control*, vol. 80, no. P1, p. 104317, 2023, doi: 10.1016/j.bspc.2022.104317.

[4] W. C. Fang, K. Y. Wang, N. Fahier, Y. L. Ho, and Y. De Huang, "Development and Validation of an EEG-Based Real-Time Emotion Recognition System Using Edge AI Computing Platform With Convolutional Neural Network System-on-Chip Design," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 9, no. 4, pp. 645–657, 2019, doi: 10.1109/JETCAS.2019.2951232.

[5] T. Wang, J. Deng, and B. He, "Classification of motor imagery EEG patterns and their topographic representation," *Annual International Conference of the IEEE Engineering in Medicine and Biology - Proceedings*, vol. 26 VI, pp. 4359–4362, 2004, doi: 10.1109/iembs.2004.1404213.

[6] S. U. Amin, M. Alsulaiman, G. Muhammad, M. A. Bencherif, and M. S. Hossain, "Multilevel Weighted Feature Fusion Using Convolutional Neural Networks for EEG Motor Imagery Classification," *IEEE Access*, vol. 7, pp. 18940–18950, 2019, doi: 10.1109/ACCESS.2019.2895688.

[7] W. Ko, J. Yoon, E. Kang, E. Jun, J. S. Choi, and H. Il Suk, "Deep recurrent spatiooral neural network for motor imagery based BCI," *2018 6th International Conference on Brain-Computer Interface, BCI 2018*, vol. 2018-Janua, pp. 1–3, 2018, doi: 10.1109/IWW-BCI.2018.8311535.

[8] M. Mohammadpour, S. M. R. Hashemi, and N. Houshmand, "Classification of EEG-based emotion for BCI applications," *7th Conference on Artificial Intelligence and Robotics, IRANOPEN 2017*, pp. 127–131, 2017, doi: 10.1109/RIOS.2017.7956455.

[9] S. Abenna, M. Nahid, H. Bouyghf, and B. Ouacha, "An enhanced motor imagery EEG signals prediction system in real-time based on delta rhythm," *Biomedical Signal Processing and Control*, vol. 79, no. October 2021, 2023, doi: 10.1016/j.bspc.2022.104210.

[10] J. Wang, W. Chen, and M. Li, "A multi-classification algorithm based on multi-domain information fusion for motor imagery BCI," *Biomedical Signal Processing and Control*, vol. 79, no. P2, p. 104252, 2023, doi: 10.1016/j.bspc.2022.104252.

[11] Y. Park and W. Chung, "BCI classification using locally generated CSP features," *2018 6th International Conference on Brain-Computer Interface, BCI 2018*, vol. 2018-Janua, no. 1, pp. 1–4, 2018, doi: 10.1109/IWW-BCI.2018.8311492.

[12] K. K. Ang, Z. Y. Chin, H. Zhang, and C. Guan, "Filter Bank Common Spatial Pattern (FBCSP) in brain-computer interface," *Proceedings of the International Joint Conference on Neural Networks*, pp. 2390–2397, 2008, doi: 10.1109/IJCNN.2008.4634130.

[13] A. Sherstinsky, "Fundamentals of Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) network," *Physica D: Nonlinear Phenomena*, vol. 404, p. 132306, 2020, doi: 10.1016/j.physd.2019.132306.

[14] B. Malpartida, "BCI Competition 2008 – Graz data set A," *数据集 2a*, pp. 1–6, 2008.

[15] K. Wang, M. Xu, S. Zhang, Y. Ke, and D. Ming, "Analysis and Classification for EEG Patterns of Force Motor Imagery Using Movement Related Cortical Potentials," *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS*, vol. 2018-July, pp. 211–214, 2018, doi: 10.1109/EMBC.2018.8512184.

[16] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, and H. Zhang, "Filter bank common spatial pattern algorithm on BCI competition IV datasets 2a and 2b," *Frontiers in Neuroscience*, no. MAR, 2012, doi: 10.3389/fnins.2012.00039.

[17] J. Xu, H. Zheng, J. Wang, D. Li, and X. Fang, "Recognition of eeg signal motor imagery intention based on deep multi-view feature learning," *Sensors (Switzerland)*, vol. 20, no. 12, pp. 1–16, 2020, doi: 10.3390/s20123496.

[18] V. Bicík, V. Golda, and P. Lisoněk, "Optico-motor coordination in intact and callostomized cats.," *Activitas Nervosa Superior*, vol. 12, no. 2, pp. 169–170, 1970.